

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

Let's solve this step by step :

- 1. **Total file size**: 200 GB
- 2. **Normal download speed**: 2 GB / minute
- 3. **40 % of the way through**:
 - 40 % of 200 GB = $0.40 \times 200 = 80$ GB
 - So , Carla downloads 80 GB before the restart .

- 4. **Time to download 80 GB at 2 GB / minute**:
 - Time = Size / Speed = $80 / 2 = 40$ minutes

- 5. **Restart due to update**:
 - Takes 20 minutes

- 6. **After restart , she has to start over**:
 - She needs to download the entire file again from the beginning , so she downloads 200 GB again at 2 GB / minute
 - Time = $200 / 2 = 100$ minutes

- 7. **Total time** = time before restart + restart time + time after restart
= $40 + 20 + 100 = 160$ minutes

So , the total time it takes to download the file is 160 minutes .

<answer> 160 </answer>

